Ex. 9.1 Q42. $a_n = \left(2 - \frac{1}{2^n}\right)\left(3 + \frac{1}{2^n}\right)$

Solution. Since $\frac{1}{2^n} \to 0$, we have

$$\lim_{n\to\infty} 2 - \frac{1}{2^n} = 2$$

and

$$\lim_{n\to\infty} 3 + \frac{1}{2^n} = 3.$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(2 - \frac{1}{2^n}\right)\lim_{n\to\infty}\left(3 + \frac{1}{2^n}\right) = 2 \cdot 3 = 6.$$

The sequence $\{a_n\}$ converges and its limit is 6.

Ex. 9.1 Q52. $a_n = \frac{3^n}{n^3}$

Solution. Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{3^x}{x^3} \quad \text{for all } x \in [1, \infty).$$

By l'Hôpital's rule,

$$\begin{aligned}
\lim_{x\to\infty} f(x) &= \lim_{x\to\infty} \frac{3^x}{x^3} \\
&= \lim_{x\to\infty} \frac{3^x \ln 3}{3x^2} \\
&= \lim_{x\to\infty} \frac{3^x (\ln 3)^2}{6x} \\
&= \lim_{x\to\infty} \frac{3^x (\ln 3)^3}{6} \\
&= \infty.
\end{aligned}$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3^n}{n^3} = \infty.$$

We conclude that the sequence $\{a_n\}$ diverges.

Ex. 9.1 Q71. $a_n = \left(\frac{1}{n}\right)^{1/(\ln n)}$

Solution. For $n \geq 2$, we have

$$a_n = \left(\frac{1}{n}\right)^{1/(\ln n)} = e^{\ln\left(\left(\frac{1}{n}\right)^{1/(\ln n)}\right)} = e^{\frac{\ln(1/n)}{\ln n}} = e^{\frac{\ln 1 - \ln n}{\ln n}} = e^{-1}.$$

Hence

$$\lim_{n\to\infty} a_n = e^{-1} = \frac{1}{e}.$$

The sequence $\{a_n\}$ converges and its limit is $\frac{1}{e}$.

Ex. 9.1 Q79. $a_n = \left(\frac{3n+1}{3n-1}\right)^n$

Solution 1. Note that

$$a_n = \left(1 + \frac{2}{3n-1}\right)^n = e^{\ln\left(1+\frac{2}{3n-1}\right)^n} = e^{n\ln\left(1+\frac{2}{3n-1}\right)}.$$

Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{\ln\left(1 + \frac{2}{3x-1}\right)}{\frac{1}{x}} \quad \text{for all } x \in [1, \infty).$$

By l'Hôpital's rule,

$$\begin{aligned}
\lim_{x\to\infty} f(x) &= \lim_{x\to\infty} \frac{\ln\left(1 + \frac{2}{3x-1}\right)}{\frac{1}{x}} \\
&= \lim_{x\to\infty} \frac{\frac{-6}{(3x-1)^2} \cdot \frac{1}{1+\frac{2}{3x-1}}}{-\frac{1}{x^2}} \\
&= \lim_{x\to\infty} \frac{6x^2}{(3x-1)^2\left(1 + \frac{2}{3x-1}\right)}
\end{aligned}$$

$$
\begin{aligned}
&= \lim_{x\to\infty} \frac{6x^2}{(9x^2-6x+1)\left(1+\frac{2}{3x-1}\right)} \\
&= \lim_{x\to\infty} \frac{6}{\left(9-\frac{6}{x}+\frac{1}{x^2}\right)\left(1+\frac{2}{3x-1}\right)} \\
&= \frac{2}{3}.
\end{aligned}
$$

Hence

$$
\lim_{n\to\infty} n \ln\left(1+\frac{2}{3n-1}\right) = \frac{2}{3}.
$$

Since e^x is continuous at $\frac{2}{3}$, we get

$$
\lim_{n\to\infty} a_n = \lim_{n\to\infty} e^{n\ln\left(1+\frac{2}{3n-1}\right)} = e^{\lim_{n\to\infty} n\ln\left(1+\frac{2}{3n-1}\right)} = e^{2/3}.
$$

The sequence $\{a_n\}$ converges and its limit is $e^{2/3}$.

Solution 2. Note that

$$a_n = \left(1 + \frac{2}{3n-1}\right)^n = \left[\left(1 + \frac{2}{3n-1}\right)^{\frac{3n-1}{2}}\right]^{\frac{2n}{3n-1}}.$$

Since

$$\lim_{n\to\infty}\left(1 + \frac{2}{3n-1}\right)^{\frac{3n-1}{2}} = e \quad \text{and} \quad \lim_{n\to\infty}\frac{2n}{3n-1} = \lim_{n\to\infty}\frac{2}{3-\frac{1}{n}} = \frac{2}{3},$$

we get

$$\lim_{n\to\infty} a_n = \left[\lim_{n\to\infty}\left(1 + \frac{2}{3n-1}\right)^{\frac{3n-1}{2}}\right]^{\lim_{n\to\infty}\frac{2n}{3n-1}} = e^{2/3}.$$

The sequence $\{a_n\}$ converges and its limit is $e^{2/3}$.

Remark. We have used two facts which are not stated in the textbook. The first is a "subsequence" of a convergent sequence has the same limit, which justify why $\lim_{n\to\infty}\left(1 + \frac{2}{3n-1}\right)^{\frac{3n-1}{2}} = \lim_{k\to\infty}\left(1 + \frac{1}{k}\right)^k$. The second is if $a_n \to a > 0$ and $b_n \to b$, then $a_n^{b_n} \to a^b$. Both of them are not stated in the textbook.

Ex. 9.1 Q95. $a_n = \frac{(\ln n)^{200}}{n}$

Solution 1. Note that

$$a_n = \frac{(\ln n)^{200}}{n} = \left(\frac{\ln n}{n^{1/200}}\right)^{200}.$$

Let $k = n^{1/200}$. Then $k \to \infty$ as $n \to \infty$, and

$$\frac{\ln n}{n^{1/200}} = \frac{\ln(k^{200})}{k} = \frac{200 \ln k}{k}.$$

Since $\lim_{k\to\infty} \frac{\ln k}{k} = 0$, we get

$$\lim_{n\to\infty} \frac{\ln n}{n^{1/200}} = \lim_{k\to\infty} \frac{200 \ln k}{k} = 0.$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{\ln n}{n^{1/200}}\right)^{200} = \left(\lim_{n\to\infty} \frac{\ln n}{n^{1/200}}\right)^{200} = 0.$$

The sequence $\{a_n\}$ converges and its limit is 0.

Solution 2. Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{(\ln x)^{200}}{x} \quad \text{for all } x \in [1, \infty).$$

Since $\lim_{x\to\infty} (\ln x)^k = \infty$ for $k = 1, \ldots, 200$ and $\lim_{x\to\infty} x = \infty$, we apply l'Hôpital's rule repeatedly to get

$$\begin{aligned}
\lim_{x\to\infty} \frac{(\ln x)^{200}}{x} &= \lim_{x\to\infty} \frac{200(\ln x)^{199}}{x} \\
&= \lim_{x\to\infty} \frac{200 \cdot 199(\ln x)^{198}}{x} \\
&\quad \vdots \\
&= \lim_{x\to\infty} \frac{200 \cdot 199 \cdots 2 \ln x}{x} = \lim_{x\to\infty} \frac{200!}{x} = 0.
\end{aligned}$$

Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{(\ln n)^{200}}{n} = 0.$$

The sequence $\{a_n\}$ converges and its limit is 0.

Ex. 9.1 Q97. $a_n = n - \sqrt{n^2 - n}$

Solution. Note that

$$\begin{aligned} a_n &= n - \sqrt{n^2 - n} \\ &= \frac{\left(n - \sqrt{n^2 - n}\right)\left(n + \sqrt{n^2 - n}\right)}{n + \sqrt{n^2 - n}} \\ &= \frac{n}{n + \sqrt{n^2 - n}} \\ &= \frac{1}{1 + \sqrt{1 - \frac{1}{n}}}. \end{aligned}$$

Since $\frac{1}{n} \to 0$, we have $\sqrt{1 - \frac{1}{n}} \to 1$. Hence

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{1 + \sqrt{1 - \frac{1}{n}}} = \frac{1}{1 + \lim_{n\to\infty} \sqrt{1 - \frac{1}{n}}} = \frac{1}{1+1} = \frac{1}{2}.$$

The sequence $\{a_n\}$ converges and its limit is $\frac{1}{2}$.

Ex. 9.2 Q13. $\sum_{n=0}^{\infty}\left(\frac{1}{2^n}+\frac{(-1)^n}{5^n}\right)$

Solution. Note that

$$\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n=\frac{1}{1-\frac{1}{2}}=2,$$

$$\sum_{n=0}^{\infty}\left(-\frac{1}{5}\right)^n=\frac{1}{1-\left(-\frac{1}{5}\right)}=\frac{5}{6}.$$

So

$$\begin{aligned}\sum_{n=0}^{\infty}\left(\frac{1}{2^n}+\frac{(-1)^n}{5^n}\right)&=\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n+\sum_{n=0}^{\infty}\left(-\frac{1}{5}\right)^n\\&=2+\frac{5}{6}\\&=\frac{17}{6}.\end{aligned}$$

The series converges and its sum is $\frac{17}{6}$.

Ex. 9.2 Q37. $\sum_{n=1}^{\infty} \ln \frac{1}{n}$

Solution. Let $a_n = \ln \frac{1}{n}$. Then

$$\begin{aligned} \lim_{n\to\infty} a_n &= \lim_{n\to\infty} (\ln 1 - \ln n) \\ &= \lim_{n\to\infty} (-\ln n) \\ &= -\infty. \end{aligned}$$

Since $\lim_{n\to\infty} a_n$ does not exist, by the n-th Term Test, the series $\sum_{n=1}^{\infty} \ln \frac{1}{n}$ diverges.

Ex. 9.2 Q42. $\sum_{n=1}^{\infty}(\tan n - \tan(n-1))$

Solution. Let $s_n = \sum_{k=1}^{n}(\tan k - \tan(k-1))$ for $n \in \mathbb{N}$. Then

$$\begin{aligned} s_n &= (\tan 1 - \tan 0) + (\tan 2 - \tan 1) + \cdots + (\tan n - \tan(n-1)) \\ &= \tan n - \tan 0 \\ &= \tan n. \end{aligned}$$

So $s_n \to \infty$. Therefore the series $\sum_{n=1}^{\infty}(\tan n - \tan(n-1))$ diverges.

Ex. 9.2 Q57. $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$

Solution. Let

$$a_n = \cos\left(\frac{n\pi}{2}\right).$$

Note that

$$a_1 = 0, \quad a_2 = -1, \quad a_3 = 0, \quad a_4 = 1, \quad a_5 = 0, \ldots$$

In general,

$$a_n = \begin{cases} 0 & \text{if } n \text{ is odd,} \\ -1 & \text{if } n = 4k+2 \text{ where } k \geq 0, \\ 1 & \text{if } n = 4k \text{ where } k \geq 1. \end{cases}$$

Hence $\{a_n\}$ diverges. Specifically, $\lim_{n\to\infty} a_n$ does not exist.

By the n-th Term Test, the series $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$ diverges.

Ex. 9.2 Q62. $\sum_{n=0}^{\infty} \frac{2}{x^n}, \quad |x| > 1$

Solution. Since $|x| > 1$, we have $\left|\frac{1}{x}\right| < 1$. Hence the series $\sum_{n=0}^{\infty} \left(\frac{1}{x}\right)^n$ converges. So

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{2}{x^n} &= 2 \sum_{n=0}^{\infty} \left(\frac{1}{x}\right)^n \\
&= 2 \cdot \frac{1}{1 - \frac{1}{x}} \\
&= \frac{2x}{x-1}.
\end{aligned}$$

The series converges and its sum is $\frac{2x}{x-1}$.

Ex. 9.2 Q76. $\sum_{n=0}^{\infty} (\ln(4e^n - 1) - \ln(2e^n + 1))$

Solution. Let

$$a_n = \ln(4e^n - 1) - \ln(2e^n + 1).$$

Then

$$a_n = \ln\left(\frac{4e^n - 1}{2e^n + 1}\right) = \ln\left(\frac{4 - e^{-n}}{2 + e^{-n}}\right).$$

Since $e^{-n} \to 0$, we have

$$\lim_{n\to\infty} \frac{4 - e^{-n}}{2 + e^{-n}} = 2.$$

Since $\ln x$ is continuous at 2, we get

$$\lim_{n\to\infty} a_n = \ln\left(\lim_{n\to\infty} \frac{4 - e^{-n}}{2 + e^{-n}}\right) = \ln 2 \neq 0.$$

By the n-th Term Test, the series $\sum_{n=0}^{\infty} (\ln(4e^n - 1) - \ln(2e^n + 1))$ diverges.

Ex. 9.3 Q33. $\sum_{n=3}^{\infty} \frac{1/n}{\ln n\sqrt{(\ln n)^2-1}}$

Solution. Let $f : [3, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{1}{x(\ln x)\sqrt{(\ln x)^2 - 1}} \quad \text{for all } x \in [3, \infty).$$

Then

$$\begin{aligned}
\int_3^{\infty} \frac{1}{x(\ln x)\sqrt{(\ln x)^2 - 1}}\, dx &= \lim_{b\to\infty} \int_3^{b} \frac{1}{x(\ln x)\sqrt{(\ln x)^2 - 1}}\, dx \\
&= \lim_{b\to\infty} \int_{\ln 3}^{\ln b} \frac{1}{u\sqrt{u^2 - 1}}\, du \qquad [u = \ln x] \\
&= \lim_{b\to\infty} \left[\sec^{-1} u\right]_{\ln 3}^{\ln b} \\
&= \lim_{b\to\infty} \left(\sec^{-1}(\ln b) - \sec^{-1}(\ln 3)\right) = \frac{\pi}{2} - \sec^{-1}(\ln 3).
\end{aligned}$$

So $\int_3^{\infty} \frac{1}{x(\ln x)\sqrt{(\ln x)^2-1}}\, dx$ converges. Since f is continuous, positive and decreasing on $[3, \infty)$, by the Integral Test, the series $\sum_{n=3}^{\infty} \frac{1}{n(\ln n)\sqrt{(\ln n)^2-1}}$ converges.

Ex. 9.3 Q38. $\sum_{n=1}^{\infty} \frac{2}{1+e^n}$

Solution. Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \frac{2}{1+e^x} \quad \text{for all } x \in [1, \infty).$$

Then

$$\begin{aligned}
\int_1^{\infty} \frac{2}{1+e^x}\, dx &= \lim_{b\to\infty} \int_1^b \frac{2}{1+e^x}\, dx \\
&= 2 \lim_{b\to\infty} \int_e^{e^b} \frac{1}{u(u+1)}\, du \qquad [u = e^x] \\
&= 2 \lim_{b\to\infty} \int_e^{e^b} \left(\frac{1}{u} - \frac{1}{u+1}\right) du \\
&= 2 \lim_{b\to\infty} \left[\ln u - \ln(u+1)\right]_e^{e^b} \\
&= 2 \lim_{b\to\infty} \left(\ln \frac{e^b}{e^b+1} - \ln \frac{e}{e+1}\right) = -2 \ln \frac{e}{e+1}.
\end{aligned}$$

So $\int_1^{\infty} \frac{2}{1+e^x}\, dx$ converges. Since f is continuous, positive and decreasing on $[1, \infty)$, by the Integral Test, the series $\sum_{n=1}^{\infty} \frac{2}{1+e^n}$ converges.

Ex. 9.3 Q46. $\sum_{n=1}^{\infty} \operatorname{sech}^2 n$

Solution. Let $f : [1, \infty) \to \mathbb{R}$ be the function defined by

$$f(x) = \operatorname{sech}^2 x \quad \text{for all } x \in [1, \infty).$$

Then

$$\begin{aligned}
\int_1^{\infty} \operatorname{sech}^2 x \, dx &= \lim_{b \to \infty} \int_1^b \operatorname{sech}^2 x \, dx \\
&= \lim_{b \to \infty} [\tanh x]_1^b \\
&= \lim_{b \to \infty} (\tanh b - \tanh 1) \\
&= 1 - \tanh 1.
\end{aligned}$$

So $\int_1^{\infty} \operatorname{sech}^2 x \, dx$ converges. Since f is continuous, positive and decreasing on $[1, \infty)$, by the Integral Test, the series $\sum_{n=1}^{\infty} \operatorname{sech}^2 n$ converges.